DGMS 2019 • SECTION K: ENGINEERING GEOLOGY & ROCK MECHANICS

Question 79: Bieniawski's Rock Mass Rating (RMR) Parameters

Official Examination Guide • Geomechanics Classification System vs. Intact Rock Weatherability Indices

OFFICIAL EXAMINATION QUESTION

Which one of the following is NOT considered for the calculation of rock mass rating?

(a) Uniaxial compressive strength	(b) Condition of discontinuities
(c) Rock quality designation	**(d) Slake durability index** — CORRECT ANSWER (NOT INCLUDED)

1. Executive Summary & Core Principle

The **Rock Mass Rating (RMR)** system, also universally known as the **Geomechanics Classification**, was developed by **Z.T. Bieniawski (1973, 1976, 1989)**. It is one of the most widely utilized empirical design tools in civil and mining engineering for assessing rock mass quality, designing tunnel support systems, and evaluating rock slope and foundation stability.

The calculation of the basic and adjusted RMR value relies strictly upon **six specific geotechnical parameters**:

1. **Uniaxial Compressive Strength (UCS)** of intact rock material (or Point Load Index $I_{s(50)}$) — Rating: 0 to 15.
2. **Rock Quality Designation (RQD)** (Deere 1964) — Rating: 3 to 20.

3. **Spacing of Discontinuities** — Rating: 5 to 20.
4. **Condition of Discontinuities** (roughness, separation/aperture, persistence, infilling/gouge, and weathering) — Rating: 0 to 30.
5. **Groundwater Conditions** (inflow rate per 10 m tunnel length, joint water pressure ratio, or moisture state) — Rating: 0 to 15.
6. **Adjustment for Discontinuity Orientation** (favorable, fair, unfavorable, or very unfavorable relative to the tunnel axis, dam foundation, or slope strike) — Adjustment: 0 to -12 (tunnels), up to -25 (foundations), up to -60 (slopes).

In contrast, the **Slake Durability Index (I_{d2})** (Franklin & Chandra 1972, ISRM 1979) measures the percentage dry weight retained by a sample of rock after two standard cycles of wetting, oven-drying, and mechanical tumbling in a slake drum. It is exclusively an index of weathering susceptibility and slaking degradation for weak, argillaceous rocks (shales, mudstones, marls). It is **NOT** an input parameter in the RMR formula.

Therefore, the parameter that is **NOT considered** in the calculation of Rock Mass Rating is **(d) Slake durability index**.



Figure 79.1: The six fundamental parameters governing Bieniawski's Rock Mass Rating (RMR) system (Geomechanics Classification) contrasted with the excluded Slake Durability Index (Grounded in Bieniawski 1989 and Hoek & Brown 1980).

2. Mathematical Formulation and Parameter Weights (Bieniawski 1989)

The total adjusted RMR score is computed as:

```
RMR = R₁ (UCS) + R₂ (RQD) + R₃ (Spacing) + R₄ (Condition) + R₅ (Groundwater) + R₆ (Orientation Adjustment)
```

Parameter	Measured Property	Rating Range	Engineering Significance
1. Strength of Intact Rock (R_1)	Point Load Index ($>10 \text{ MPa}$) or Uniaxial Compressive Strength ($>250 \text{ MPa}$)	0 – 15	Evaluates intact rock block crushing resistance under applied overburden stresses.
2. Rock Quality Designation (R_2)	Percentage of intact core pieces $\ge 10\text{ cm}$ in total drill core run	3 – 20	Quantifies rock mass fracturing frequency; $\text{RQD} = 90\text{--}100\%$ gives maximum rating of 20.
3. Spacing of Discontinuities (R_3)	Perpendicular distance between joints ($>2\text{ m}$ to $<60\text{ mm}$)	5 – 20	Dictates the size of individual interlocking rock blocks within the rock mass.
4. Condition of Discontinuities (R_4)	Persistence, aperture, roughness, infilling thickness/	0 – 30	Highest single weighted parameter; controls shear strength along

Parameter	Measured Property	Rating Range	Engineering Significance
	type, joint wall weathering		potential failure surfaces.
5. Groundwater Conditions (R_5)	Completely dry, damp, wet, dripping, or flowing water; or pore pressure ratio	0 – 15	Quantifies effective stress reduction and lubrication along joint planes.
6. Orientation Adjustment (R_6)	Strike and dip orientation relative to engineering excavation	0 to -12 (Tunnels) 0 to -25 (Foundations)	Penalizes unfavorable daylighting of discontinuities dipping into excavations.

3. Rock Mass Classes Derived from Total RMR

RMR Value	Rock Class	Rock Quality Description	Cohesion (c)	Friction Angle (ϕ)	Stand-up Time
81 – 100	Class I	Very Good Rock	$> 400\text{ kPa}$	$> 45^\circ$	20 years for 15 m span
61 – 80	Class II	Good Rock	$300 – 400\text{ kPa}$	$35^\circ – 45^\circ$	1 year for 10 m span
41 – 60	Class III	Fair Rock	$200 – 300\text{ kPa}$	$25^\circ – 35^\circ$	1 week for 5 m span
		Poor Rock			

RMR Value	Rock Class	Rock Quality Description	Cohesion (c)	Friction Angle (ϕ)	Stand-up Time
21 – 40	Class IV		$100 – 200\text{ kPa}$	$15^\circ – 25^\circ$	10 hours for 2.5 m span
< 20	Class V	Very Poor Rock	$< 100\text{ kPa}$	$< 15^\circ$	30 minutes for 1 m span

4. Systematic Option Evaluation

- **(a) Uniaxial compressive strength: CONSIDERED.** Parameter 1 (R_1) directly evaluates UCS (or Point Load Index), with ratings from 0 to 15.
- **(b) Condition of discontinuities: CONSIDERED.** Parameter 4 (R_4) evaluates discontinuity roughness, infilling, and weathering, carrying the largest weight (up to 30 points).
- **(c) Rock quality designation: CONSIDERED.** Parameter 2 (R_2) directly uses Deere's RQD, with ratings from 3 to 20.
- **(d) Slake durability index: NOT CONSIDERED (CORRECT ANSWER).** Slake durability (I_{d2}) is an independent index for shale and mudrock weathering in laboratory drums; it is absent from the RMR framework.

> **Exam Tip:** RMR = UCS + RQD + Spacing + Condition + Groundwater ± Orientation. Slake durability is NOT in RMR! Hence, **(d)** is the correct answer.

← Previous: Q.78 Indian Hydrocarbon Deposits **Master Portal Index** **Next: Q.80 Dam Foundation Geological Conditions →**